

South-Facing Solar Tracker Performance on a North Roof

MPP Characterization · Linear Actuator Tracking · INA219 Power Monitoring via ESP8266 | Department of Engineering | Spring 2025

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INTRODUCTION & BACKGROUND

Solar Energy & MPP

Solar PV panels produce maximum power only at the Maximum Power Point (MPP) - a specific voltage-current balance that shifts with irradiance and temperature. Identifying and operating at MPP is critical to maximizing energy harvest.

Site & Resource Data (Eastern Pennsylvania, 41°N)

Long-term solar resource data (NREL) shows:

- Fixed tilt: 4.1 peak sun hours/day
- 1-axis tracking: 5.3 peak sun hours/day

North-Facing Roof Challenge

A north-facing panel receives 45% less power than a south-facing panel. This reduces the peak performance to 8.25 W. This significant loss motivates the development of an active optimization system.

Research Approach

MPP is characterized using a 0-300 Ω resistive sweep with real-time INA219 power monitoring. A custom 3D-printed linear actuator then dynamically orients the panel for peak irradiance.

METHODS & MATERIALS

Monocrystalline Solar Panel

15W Monocrystalline, 36 cells, 20.5% efficiency

ESP8266

WiFi microcontroller, live webpage server

INA219 Breakout

I2C power monitor - V, I, P in real time

0-300 Ω Rheostat

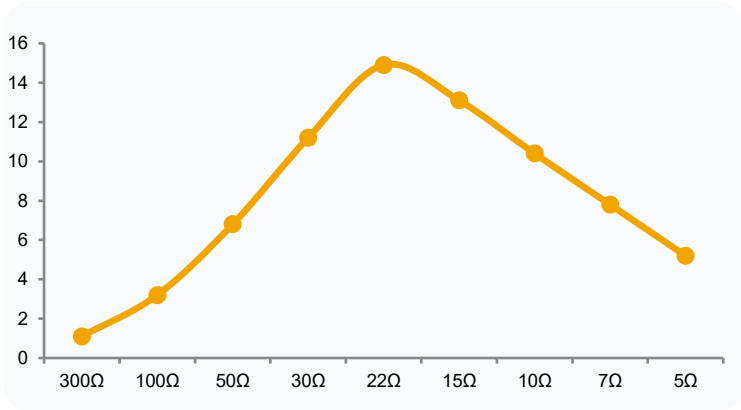
Variable load for MPP resistance sweep

3D-Printed Actuator

Custom linear actuator for panel tracking

MPP CHARACTERIZATION RESULTS

Power vs. Load Resistance



14.9 W

Peak Power

22 Ω

Optimal Load

18.2 V

MPP Voltage

TRACKING SYSTEM PERFORMANCE ANALYSIS

SYSTEM DESIGN

North-Roof Tracking Concept:

Conventional solar installations on north-facing roofs are avoided due to significantly reduced irradiance. This system transforms a low-yield surface into a productive solar asset using a motorized tracking mechanism.

Linear Actuator:

A custom-designed, 3D-printed linear actuator adjusts panel angle throughout the day. The mechanism mounts on the north-facing roof slope.

How It Works:

- Retracted position: Panel lies flat
- Extended position: Panel rotates to face south
- Tracks solar noon for maximum irradiance
- Returns to flat overnight
- Snow mode: actuator shakes the panel to shed snow accumulation

Key Innovation:

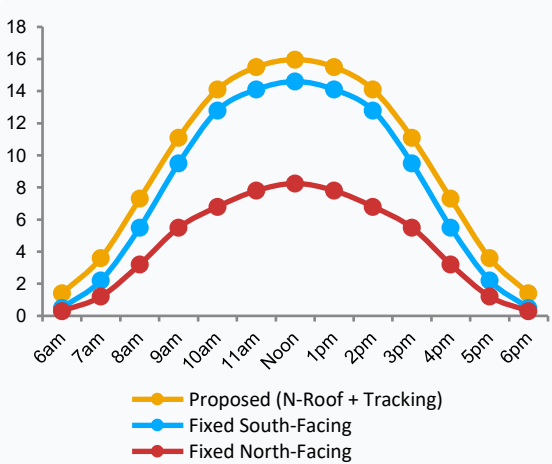
Effectively converts a north-facing surface into a south-facing one during peak hours, maximizing generation from an otherwise underutilized surface. The actuator also enables active snow shedding - a significant advantage in northeastern climates where snow accumulation can eliminate solar production for days at a time.

Actuator Power Draw:

6W peak draw
Operates 30 - 60 sec/day maximum
Total daily consumption: <0.1 Wh

DAILY POWER OUTPUT COMPARISON

Averaged Power output (W) across daylight hours

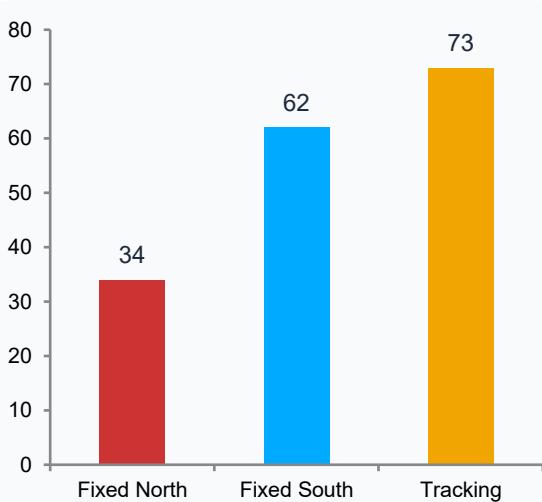


Configuration	Morning 8-10am	Solar Noon 11am-1pm	Afternoon 2-4pm
Tracking	8 -14 W	14.8-15.1 W	8-14 W
Fixed South	7.5-13 W	14.1-14.6 W	7.5-13 W
Fixed North	1-2.4 W	2.9-3.1 W	1-2.4 W

Key Insight: The tracking system produces ~17.5% more daily energy than a fixed south-facing panel and 111.5% more than a fixed north-facing panel.

ENERGY OUTPUT ANALYSIS

Estimated daily energy production (Wh)



33.8 Wh

Fixed North (baseline)

61.5 Wh

Fixed South (reference)

73 Wh

Tracking System

<0.1 Wh

Actuator consumption

CONCLUSIONS & FUTURE WORK

System Performance:

The tracking system produces 73 Wh/day vs 62 Wh/day for a fixed south-facing setup, approximately 17.5%.

The proposed system elevates a north-facing installation to a south-facing one, effectively converting a low-yield surface into a productive energy source.

Actuator ROI:

The actuator uses 6 W for 30 - 60 s/day, which is 0.05-0.1 Wh/day (<0.2%) of the total output

Energy gain is ~17.5% over fixed tilt. Added cost is \$120-\$200 (actuator + brackets). Estimated payback is 4-9 years, depending on electricity prices and construction costs.

Design Justification:

Single-axis tracking increases output by maintaining optimal sun angle.

Gain is driven by higher effective peak sun hours (5.3 PSH vs 4.1 PSH). The system improves the energy yield without changing panel size or location.

Future Work:

Battery storage integration - Seasonal performance analysis